

Assignment-1 (02-01-2023)

- 1) Switch from Notepad++ to vs code IDE.
- 2) VS code overview & Installation
- 3) Basic Practice of HTML source code in VS code.

Assignment-2 (09-01-2023)

- 1) Basic example of css practice in all types of css



WIRELESS HEADPHONES

THINKING OF BUYING A HEADPHONE?

HTML Business Templates

Shopping Now

Find a quality and right headphon

Use marquee tag here

III. HIGH QUALITY

- ORIGINAL DEVICES
- YEAR WARRANTY
- WEEK TO CHECK

II. Visit us and see yourself!

- Soft Skills
- Business Skills
- Presentaion

I. About us

- Contact

"Success victory"

Life is simple.... Trick to win

Are you want that???

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit.

[Click Here to get more](#)

Email-Id:

First Name:

Password:

Copyright @Your student ID

Assignment – 3 (16-01-2023)



OUR FOCUS IS OUR CLIENT SUCCESS!

"Finance Buzz"

Understated Elegance... Unquestionable Taste... Uncompromising Quality...

ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

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Inquiry Form

Email-Id:

First Name:

Password:

Copyright@Your student ID

Assignment-4 (13-02-2023 , 20-02-2023)

Cascading Style Sheet (CSS)

1. Create the following HTML Web page using CSS Internal Style.

This is a Heading

This is a paragraph.

This is another paragraph.

2. Create following HTML Web Page using CSS External file.

THIS IS A HEADING

Lorem Ipsum Dolor Sit Amet, Consectetur Adipiscing Elit, Sed Do Eiusmod Tempor Incidunt Ut Labore Et Dolore Magna Aliqua. Ut Enim Ad Minim Veniam, Quis Nostrud Exercitation Ullamco Laboris Nisi Ut Aliquip Ex Ea Commoda Consequat. Duis Aute Inure Dolor In Reprehenderit In Voluptate Velit Esse Cillum Dolore Eu Fugiat Nulla Pariatur. Excepteur Sint Occaecat Cupidatat Non Proident, Sunt In Culpa Qui Officia Deserunt Mollit Anim Id Est Laborum

3. Create Following Web page to display images using background properties.



4. Create HTML file to design following web page using style.

Lab Assignment || BCA || SEM II || CA128 : Advance Web Designing

Hello World!

Smt. Chandaben Mohanbhai Patel Institute of Computer Applications, is a constituent post-graduate institution of CHARUSAT offering a three-year full-time Master of Computer Applications (MCA) programme with the intake of 180 students per year, a two-year full-time MCA Lateral programme with an intake of 60 students per year, a two-year full-time Master of Science in Information Technology (M.Sc.(IT)) programme with intake of 60 students per year, a five-year full-time Dual Degree Master of Computer Applications (MCA-Dual Degree BCA + MCA) programme with intake of 60 students per year and a Doctor of Philosophy (Ph. D.) programme (Full time/Part time).

The current scenario of industrial and governmental developments in the country provides an ever growing need for highly trained manpower in the field of Computer Applications and Information Technology.

- Coffee
- Tea
 - Black tea
 - Green tea
- Milk

study

the devotion of time and attention to acquiring knowledge on an academic subject, esp. by means of books

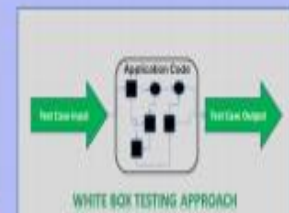
design

a plan or drawing produced to show the look and function or workings of a building, garment, or other object before it is built or made purpose, planning, or intention that exists or is thought to exist behind an action, fact, or material object

business

work

a person's regular occupation, profession, or trade



5. Create following web page using HTML and CSS.

The Bookworm: A Better Book Store

"Old books are best--how tale and rhyme
Float with us down the stream of time!"
-- Clarence Urry, *Old Songs are Best*

The Bookworm Bookshop
1345 Applewood
Springfield, CA 94325
(415) 555-0034

Contents

- [About the Bookworm Bookshop](#)
- [Recent Titles](#)
- [Upcoming Events](#)

About the Bookworm Bookshop

Since 1933, The Bookworm Bookshop has offered rare and hard-to-find titles for the discerning reader. The Bookworm offers:

- Friendly, knowledgeable, and courteous help
- Free coffee and juice for our customers
- A well-lit reading room so you can "try before you buy"
- Four friendly cats: Esmerelda, Catherine, Dulcinea and Beatrice

Our hours are 10am to 9pm weekdays, noon to 7 on weekends.

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Recent Titles (as of July 11, 2012)

- Sandra Bellweather, [Exercise 5 \(Practical List -3\)](#)
- Johnathan Tin, [Exercise 12 \(Practical List -2\)](#)
- Maxwell Burgess, [Image Map](#)
- Alison Caine, [Tables](#)

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Upcoming Events

The Wednesday Evening Book Review
meets, appropriately, on Wednesday evenings at 7 pm for coffee and a round-table discussion. Call the Bookworm for information on joining the group.

The Children's Hour
happens every Saturday at 1 pm and includes reading, games, and other activities. Cookies and milk are served.

Carole Fenney
will be at the Bookworm on Sunday, January 19, to read from her book of poems *Spiders in the Web*.

The Bookworm will be closed
March 1st to remove a family of bats that has nested in the tower. We like the company, but not the mess they leave behind!

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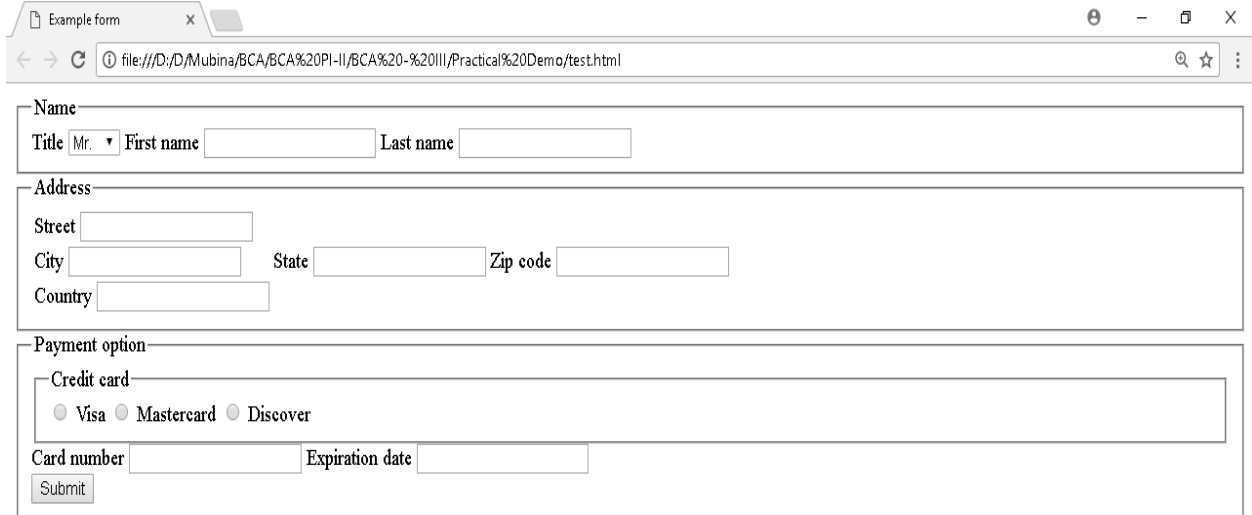
Last Updated: July 11, 2012
Webmaster: Laura Lemay lemay@bookworm.com
© copyright 2012 the Bookworm



Background Color - #CCFFCC

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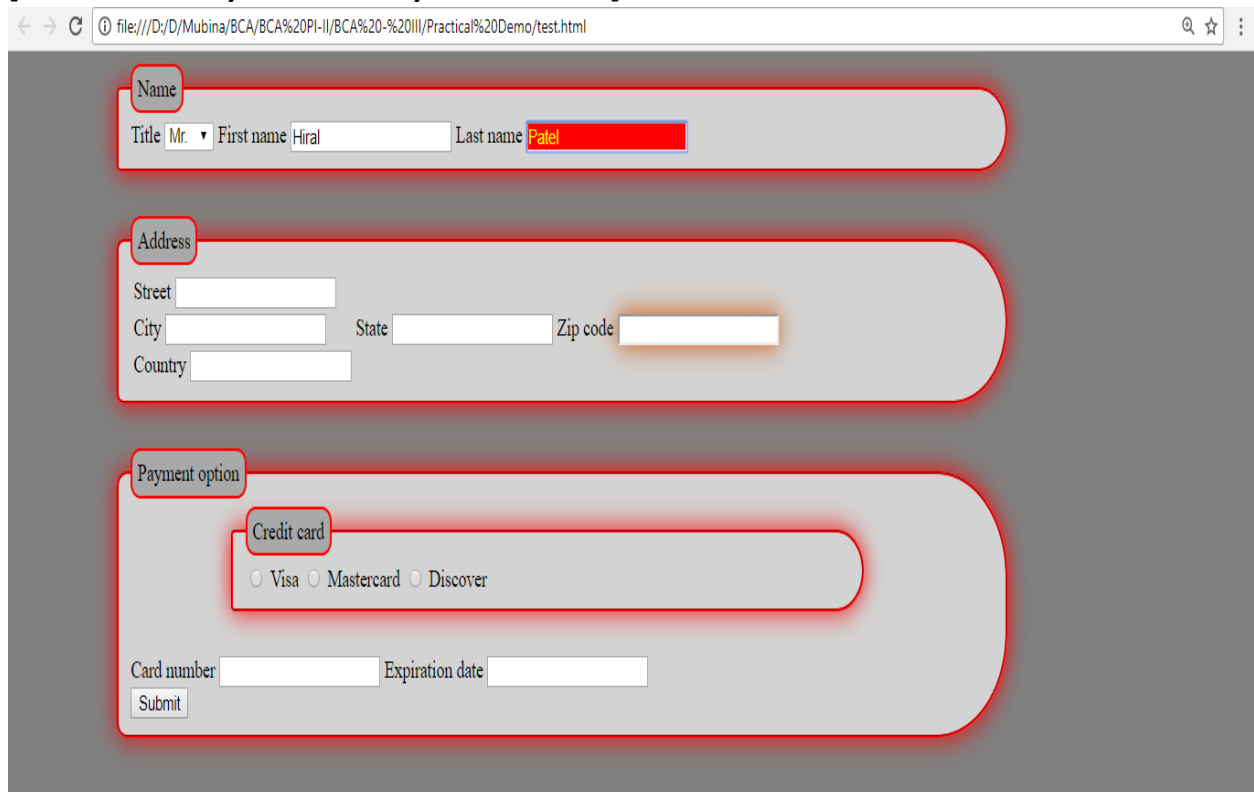
6. Create a HTML web page to design following form.



The screenshot shows a web browser window with the address bar displaying a file path. The form is unstyled and consists of the following sections:

- Name:** A dropdown menu for "Title" (set to "Mr."), and text input fields for "First name" and "Last name".
- Address:** Text input fields for "Street", "City", "State", "Zip code", and "Country".
- Payment option:** A section containing a "Credit card" label, three radio buttons for "Visa", "Mastercard", and "Discover", and text input fields for "Card number" and "Expiration date". A "Submit" button is at the bottom.

[Use external style sheet to style above form]



The screenshot shows the same web browser window, but the form is now styled with a gray background and rounded corners. The styling includes:

- Body background color:** Gray.
- Fieldset background color:** #D3D3D3.
- Legend background color:** #A9A9A9.

The form fields are now visually grouped and highlighted with red borders and rounded corners, matching the design requirements.

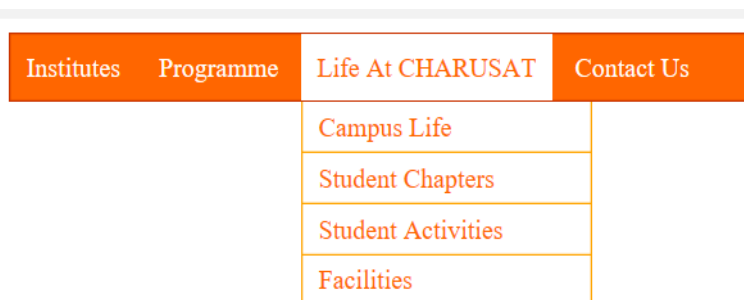
- Body background color - gray
- Fieldset background color - #D3D3D3
- Legend background color - #A9A9A9

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- Mouse over shadow color of an input element- #D35400
- On focus input element background color – red
- On focus Input element text color – yellow;



7. Create Horizontal Menu using HTML & CSS.



Institute: CMPICA, PDPIAS, I2IM,CSPIT,RPCP,MTIN,ARIP

Programme: Undergraduate,Postgraduate,Diploma/Certificate,Research

8. Create Vertical Menu using HTML & CSS.
9. Create following template using HTML and CSS



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Note:

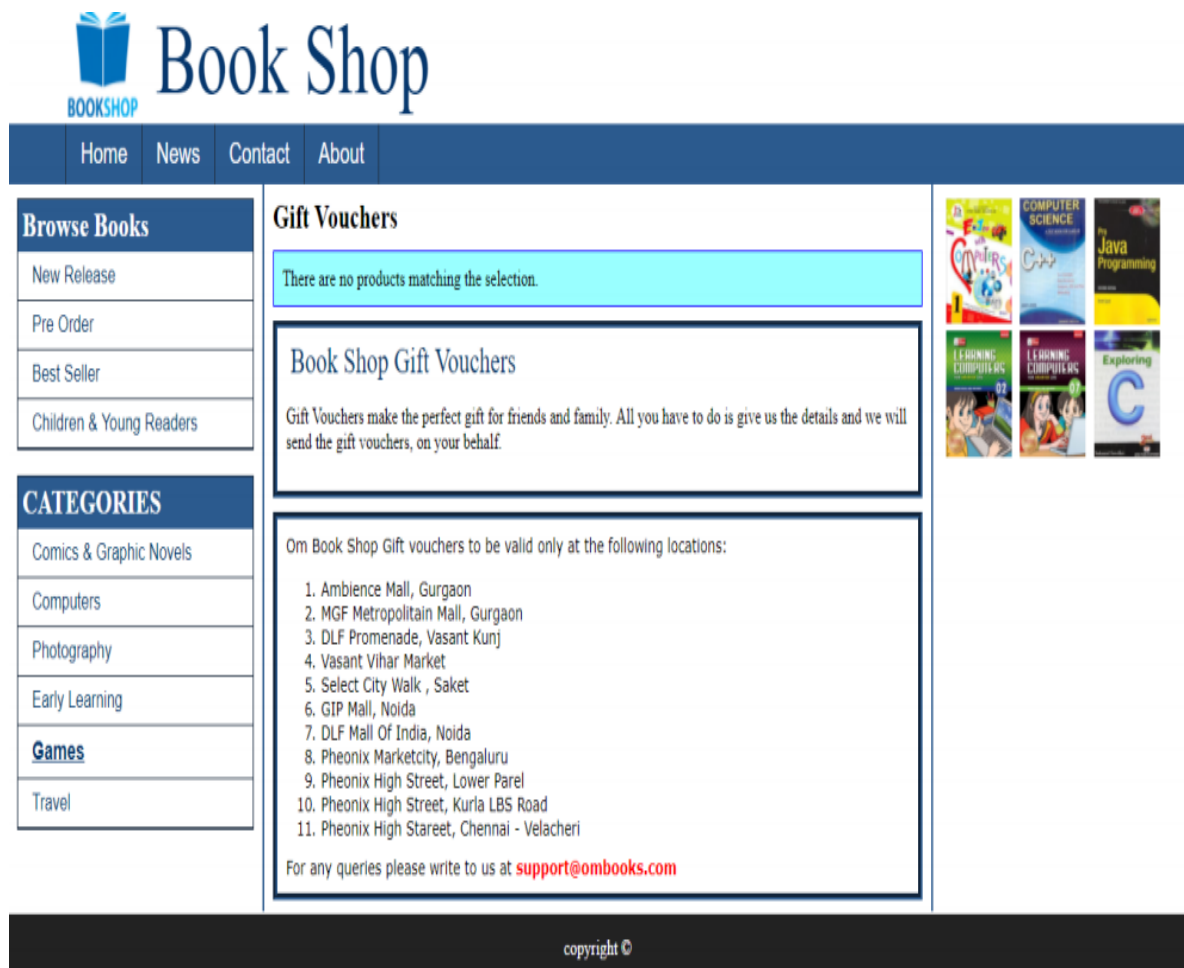
The colours used are as follows:

- Text: #330000
- Body background: #ffffcc
- Header and footer background: #ccaa66
- Left (navigation) column background: #E8D882
- Right (main content) column background: #f1e8b0

The text is displayed in Verdana font.

You should write an external/Internal CSS file. You need to modify the original HTML file only to link to this style sheet, mark up sections (using <div> tags) and set appropriate class and id attributes.

10. Create the following page using HTML & CSS:



The screenshot shows a web page for 'Book Shop'. The header includes a logo and the text 'Book Shop'. Below the header is a navigation bar with links: Home, News, Contact, About. The main content area is divided into three columns. The left column contains a 'Browse Books' section with links: New Release, Pre Order, Best Seller, Children & Young Readers. Below this is a 'CATEGORIES' section with links: Comics & Graphic Novels, Computers, Photography, Early Learning, Games, Travel. The middle column contains a 'Gift Vouchers' section. It starts with a message: 'There are no products matching the selection.' Below this is a section titled 'Book Shop Gift Vouchers' with a description: 'Gift Vouchers make the perfect gift for friends and family. All you have to do is give us the details and we will send the gift vouchers, on your behalf.' Below this is a list of locations where the vouchers are valid: 1. Ambience Mall, Gurgaon, 2. MGF Metropolitan Mall, Gurgaon, 3. DLF Promenade, Vasant Kunj, 4. Vasant Vihar Market, 5. Select City Walk, Saket, 6. GIP Mall, Noida, 7. DLF Mall Of India, Noida, 8. Pheonix Marketcity, Bengaluru, 9. Pheonix High Street, Lower Parel, 10. Pheonix High Street, Kurla LBS Road, 11. Pheonix High Stareet, Chennai - Velacheri. At the bottom of this section is a link: 'For any queries please write to us at support@ombooks.com'. The right column contains a grid of book covers, including 'COMPUTER SCIENCE', 'Java Programming', 'LEARNING COMPUTERS', and 'Exploring'. The footer contains the text 'copyright ©'.

The colours used are as follows:

Header text :- 06376D

Footer background:- # 222222



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menu background:- 2B5A8E

menu item border:- 213042 menu

font:- Arial, Helvetica, sans-serif active

menu item:- DCE5EE

menu hover:- 2B5A8E

aside border:- 2B5A8E

aside heading:- 2B5A8E

aside list:- 06376D

aside list font:- Arial, Helvetica, sans-serif gift voucher

first article background:- 99FFFF

article border:- 2B5A8E

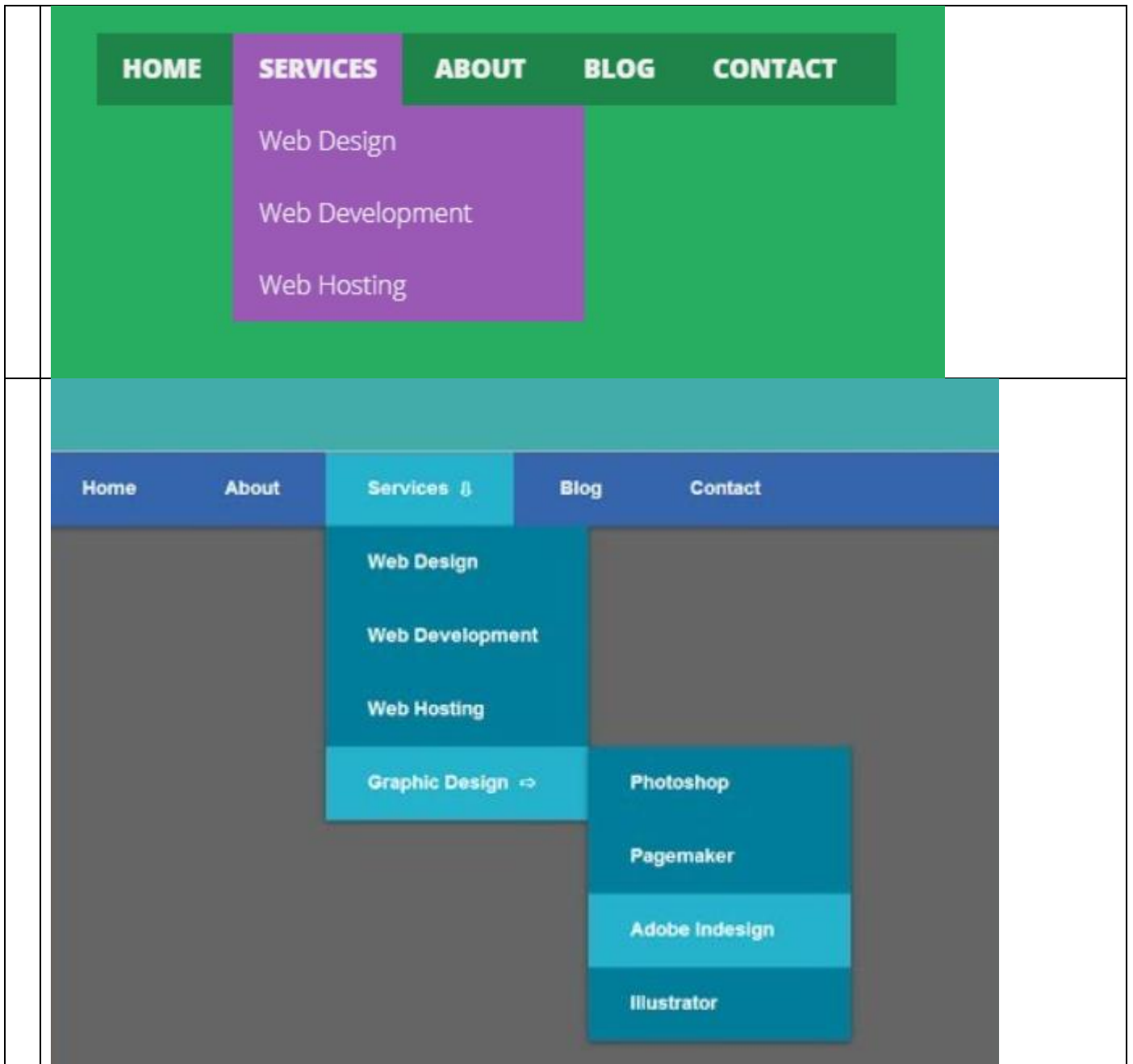
3rd article font:- Verdana, Arial, Helvetica, sans-serif

Note:-

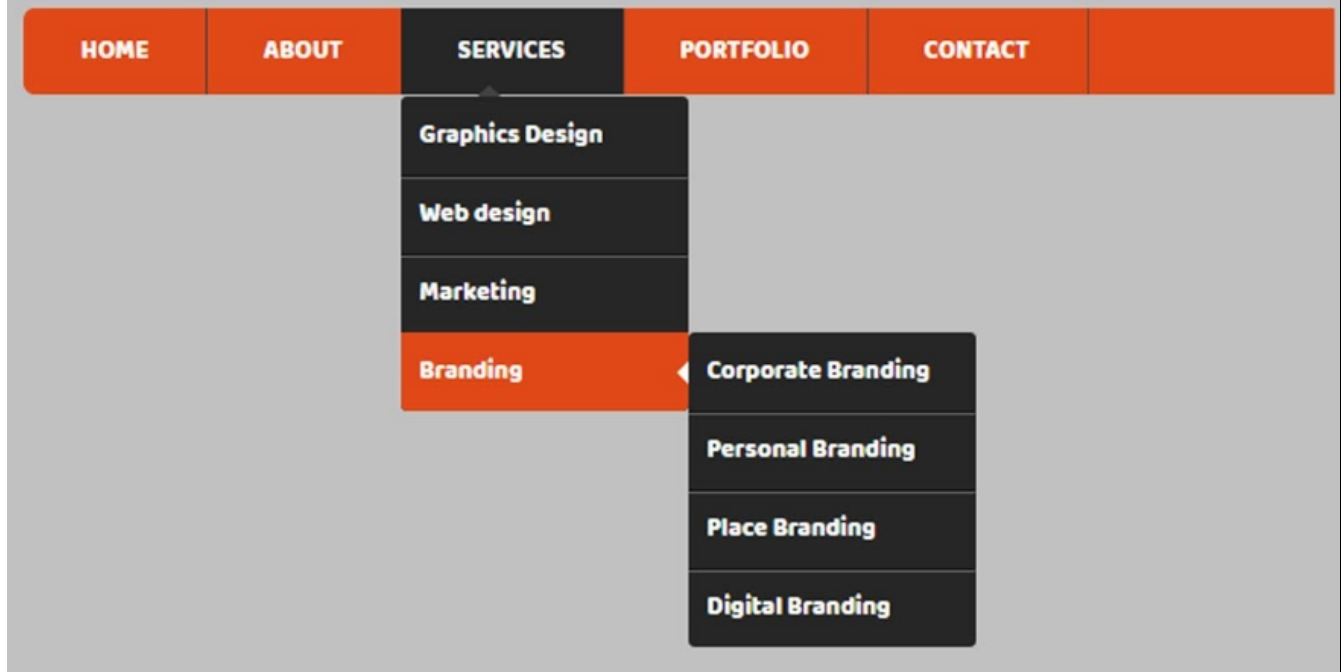
- When user move mouse cursor on aside list item then particular list item display with underline and increase font size
- When user move mouse cursor on image then image should be larger and display border



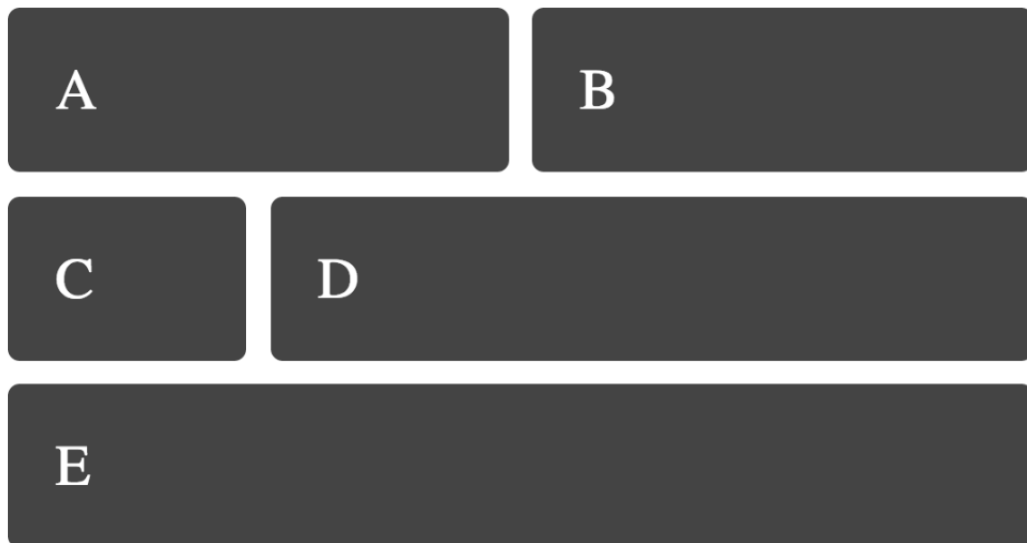
Assignment – 5 (27-02-2023)



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1 Create a webpage that prints below with the use of grid



2 Create a webpage that prints below with the use of grid



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	A	B	
	C	D	

3 Create a webpage that prints below with the use of grid

	A	B	D	
	E	C		
	F			



Lab Assignment || BCA || SEM IV || CA217 : Multi Paradigm Programming Language

1. WAP to calculate simple interest.
 $SI = P \times R \times N / 100$ (P=principal amount, R=rate of interest, N=numbers of years)
2. WAP to swap the value of two variables.
3. WAP to find largest and smallest value from given three numbers.
4. WAP to find largest odd number from three numbers and if no number is odd then print message accordingly.
5. Take three subject marks as user entry and then calculate percentage and display the class, "Distinction", "First", "Second" and so on.
6. Check whether the entered number is positive or negative.
7. Calculate area of triangle.
8. Calculate area of circle.
9. Write a program to convert Fahrenheit to Celsius.
10. Write a Python program to add 'ing' at the end of a given string (length should be at least 3). If the given string already ends with 'ing' then add 'ly' instead. If the string length of the given string is less than 3, leave it unchanged.
11. Check whether two strings are anagram or not. Hint: "listen" = "silent"
12. Take two strings from user and print which one is larger without using built-in functions.
13. Count numbers of vowels present in entered string.
14. Write a Python program to get a string made of the first 2 and the last 2 chars from a given a string. If the string length is less than 2, return instead of the empty string.
15. Write a Python program to get a string from a given string where all occurrences of its first char have been changed to '\$', except the first char itself.
Sample String : 'restart'
Expected Result : 'resta\$t'
16. Write a Python program to get a single string from two given strings, separated by a space and swap the first two characters of each string.
17. Print Fibonacci series for entered number of terms.
18. Print factorial of an entered number when number is positive integer and print appropriate message in case of negative integer.
19. Implements string, Tuple, List built-in functions.
20. Write a Python program to sum all the items in a list
21. Write a Python program to multiplies all the items in a list.
22. Write a Python program to get the largest number from a list.
23. Write a Python program to get the smallest number from a list.
24. Count numbers of vowels present in entered string using SET in python.
25. Calculate the NetSalary for the employees for different roles.
Take User input for type of employee, then respectively calculate net salary using following equation: $NetSalary = Basic + HRA(5\% \text{ of Basic}) + DA(105\% \text{ of Basic}) - PF(10\% \text{ of Basic})$.
The different roles are: Manager, Project Leader, Developer.
26. WAMP to check whether given number is EVEN/ODD, PRIME, PALINFROME, ARMSTRONG (sum of the cubes of its digits is equal to the **number** itself. E.g. 371) KRISHNAMURTHY (A **Krishnamurthy number** is a **number** whose sum of the factorial of digits is equal to the **number** itself. E.g 145)
Expected Output:



Lab Assignment || BCA || SEM IV || CA217 : Multi Paradigm Programming Language

Enter Number: 25

Menu

1. Even /Odd
2. Prime
3. Palindrome
4. Armstrongs
5. Krishnamurthy

Enter Choice: 1

27. Write a python program that will take string from user and count each letter occurrence from it.
28. Write a python program that will create a string list dynamically and then calculate length of each string items from list and create a new list of length.
29. Write a python program that will print multiplication table of entered number.
30. Create a following menu.
 - a. Credit Amount
 - b. Debit Amount
 - c. Show Balance

If user try to withdraw amount that will make balance $\leq$ minimum balance, then do not allow such transaction and print appropriate message.

31. Create following menu:
 - a. Reverse the string
 - b. Find Specific Character Occurrence
 - c. Replace the original string with sub string
32. Create following menu:
 - a. Add an element to tuple
 - b. Count particular element in tuple
 - c. Reverse Tuple
33. Create a list of name and another which contains corresponding length of that name. If corresponding length is correct than put true otherwise false.
Example:
Name = ['ankita', 'rina', 'madhuri', 'shilpa']
Ln = [5, 2, 6, 6]
Output = [False, False, True, True]
34. Create 2 list of same size. Edit it with dictionary as shown in example.

L1 = [1, 2, 3]

L2 = ["abc", "pqr", "cde"]

D1 = {1: abc, 2: pqr, 3: cde}

35. Merge two list of different size and avoid duplications, as per example.
L1 = [22, 4, 21, 5, 32]



Lab Assignment || BCA || SEM IV || CA217 : Multi Paradigm Programming Language

L2 = [2,32,4,1]

Ans = [22,2,4,32,21,1,5,32]

36. Take one tuple of any size and split it into two lists in even index and odd index. Ignore index 0.

37. Take one list and map it into the dictionary as per the example.

Lst = [1,2,3,4,5,6,7,8]

Dict = {1:[1,2],3:[3,4],5:[5,6],7:[7,8]}

38. Ask one number from the user. Ask that many string from the user. And enter this string and length of this string in the list. See the example,

Enter number: 3

Enter string : rajan

Enter string: kirtan

Enter string: yuvraj

Final list would be : [rajan,5,kirtan,6,yuvraj,6]

39. Perform above 38 assignments by defining user defined functions in Python.

40. Define a Python function "descending(l)" that returns True if each element in its input list is at most as big as the one before it

For instance:

```
>>> descending([])
```

```
True
```

```
>>> descending([4,4,3])
```

```
True
```

```
>>> descending([19,17,18,7])
```

```
False
```

41. Define a Python function "alternating(l)" that returns True if the values in the input list alternately go up and down (in a strict manner).

For instance:

```
>>> alternating([])
```

```
True
```

```
>>> alternating([1,3,2,3,1,5])
```

```
True
```

```
>>> alternating([3,2,3,1,5])
```

```
True
```

```
>>> alternating([3,2,2,1,5])
```



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False

```
>>> alternating([3,2,1,3,5])
```

False

42. Write Python program to search an element from list using function. (linear search)

Def Linearsearch(lst1,ele), if an element found return its index value and if element not found then return -1.

43. Create a user defined list of strings and search a string in that list.

44. Write a python function that checks whether a given number is in range or not.

For instance num_check(num,low,high).

45. Write a python function that returns a lesser of two given numbers if both numbers are even but returns the greater if one or both numbers are odd.

46. Write a function takes a two word string and returns true if both words begin with same letter.

47. Given two integers return true if sum of integers are true, or if one of the integer is 20. If not return false.

48. Given sentence return a sentence with word reverse.

49. Write a function that accepts a string and calculate the number of upper case letters and lower case letters.

50. Create a class to enter 3 different values and find out largest of three out of them.

51. Create a class which as 2 lists as a member of them. Print the list whose sum is higher than another.

52. Create a class which ask a number from the user. If the number is even then find factorial of it and if odd then print a series from that number to 1.

Example: if input is 4 then ans is 24 but if the number is 7 then answer is 7 6 5 4 3 2 1

53. Create a class with following fields. ID, Name, Sub1, Sub2, Total and percentage. Id, name, sub1 and sub2 will be entered by the user, calculate total and percentage by your own.

54. Create a class which ask one number from the user and enter that many members in list. Now if the list has not duplicate values then return true otherwise return false.

55. Define a Super class Account with methods interest(), deposit() and withdraw().

- interest() method will calculate interest based on the available balance in the account (for saving account interest is 10% and for current account 15%) .
- deposit() method will perform the deposit operation.
- withdraw() method will perform the withdraw operation and also put a check for minimum balance, if someone try to withdraw beyond some minimum balance then this method should print appropriate message.

** minimum balance in case of saving account is Rs.1000 and Rs. 10000 is for current type of account.



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Define two subclasses from Account class SavingAccount and CurrentAccount to inherit the Account class properties.

For above operations create following menu:

Enter the type of Account:

** SA for Saving Account Operation
**CA for Current Account Operation

1. Deposit
2. Withdraw
3. Calculate Rate of Interest

Enter the choice for operation:

56. Write a Python program to create following menu and execute its implementations as per the choice of user:

- a. Find number is Odd or Even:
- b. Find number is palindrome or not:

57. Create a class Student , with following methods:

- a. getDetails() method will take user input for getting student personal information, Name, City.
 - b. viewDetails() methods will display all information of student.
- Inherit Student class into Result class, is having following methods
- getMarks() method will take user input for 3 subject marks.
 - calResult() method will calculate percentage and find grad of student.
 - viewDetails() – Override this method to display result of student.

58. Create a class Line. When the class is instantiated, it initializes with line's two coordinate values, (x1,x2) and (y1,y2).

Create a class Line. When the class is instantiated, it initializes with line's two coordinate values, (x1,x2) and (y1,y2).

Define two methods distance() and slope() in the class.

Sample output:

Cor1=(3,2)

Cor2=(8,10)

Obj1=Line(cor1,cor2)

Obj1.distance() //9.433, $c = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2}$



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Obj1.slope() //1.6, $\text{slope} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$



Smt. Chandaben Mohanbhai Patel Institute of Computer Applications
BCA – Semester IV
CA221 – Data Structures and Algorithms

Practical Set – I
Programs on Array and Linked List

1	Write a program to accept n elements and print it.
2	Write a program to accept n elements and find the summation and average of it.
3	Write a program to accept n elements and print all odd elements and count odd elements.
4	Write a program to accept n elements and find the maximum and minimum element from it.
5	Write a program to perform the following operations on a stack. (Implement the stack using array and linked list both) • PUSH • POP • ISEMPY • ISFULL • PEEP
6	Write a program to perform the following operation on a simple queue. (Implement the queue using array) • Insert an element • Remove an element.
7	Write a program to perform the following operation on a circular queue. (implement the queue using array) • Insert an element • Remove an element
8	Write a program to create a Singly Linked List in LIFO fashion.
9	Write a program to create a Singly Linked List in FIFO fashion.
10	Write a program to create a sorted Singly Linked List.
11	Write program perform the following operations on a Singly Linked List. 1. Insert an element 2. Delete an element 3. Find the sum of elements of the List



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	- Count number of the nodes in the Linked List - Search a given element in the Linked List. - Reverse the Linked List. - Make a copy of the given Linked List - Concatenate two Linked List - Merge two Linked List. - Find the union of the two given Linked List - Find the intersection of the two given Linked List.
12	Write a program to create a sorted Doubly Linked List.
13	Write a program to create a Doubly Linked List in LIFO fashion.
14	Write a program to create a Doubly Linked List in FIFO fashion.
15	Write a program perform the following operations on a doubly Linked List. - Insert an element - Delete an element - Find the sum of element of the List - Count number of the nodes in the Linked List - Search a given element in the Linked List. - Reverse the Linked List - Make a copy of the given Linked List - Concatenate two Linked listed. - Merge two Linked List. - Find the union of the two given Linked List. - Find the intersection of the two given Linked List.
16	Write a program to create a Circular Single Linked List.
17	Write a program to create a Double Linked List.



Practical Set – II

Programs on Binary Tree

1	Write a program to create a binary search tree.
2	Write a program to create a binary search tree and print its elements in inorder. (write iterative code)
3	Write a program to create a binary search tree and print its elements in preorder. (write iterative code)
4	Write a program to create a binary search tree and print its elements in postorder. (write iterative code)
5	Write a program to make another copy of a given binary search tree.
6	Write a program to count no of leaf nodes in a given binary tree.
7	Write a program to search an element in a given binary search tree.
8	Write a program to accept n elements and check particular element is present how many times. 1. Linear Search 2. Binary Search
9	Write a program to accept n elements and arrange all the elements in Ascending order. 1. Bubble Sort 2. Selection Sort 3. Insertion Sort 4. Shell Sort 5. Quick Sort 6. Merge Sort 7. Radix Sort 8. Heap Sort 9. Topological Sort



ASSIGNMENT-1

Objectives

- Hands on with Basics of Data Definition Language (DDL) commands Like Create Table, Alter Table, Truncate Table, Drop table etc.
- Hands on with Basics of Data Manipulation Language (DML).
- To understand and apply constraints to the tables.

Create Following tables

1. Table1 : stud_master

Column Name	Datatype and Size	Constraints	Description
Sid	varchar(10)	Primary Key. Should contain BCA or MCA	Student ID. Example:21BCA001
Fname	Varchar2(50)		Name of a student. Example: Amit
Lname	Varchar2(50)		Surname of a student. Example: Patel
Branch	varchar2(10)		
Semester	Char(1)		
City	Varchar2(30)		
DOB	DATE		Format: DD-MMM-YYYY. For example: 18-JAN-2003
Gender	CHAR(1)	Should Contain M or m or F or f	

Insert following data in above table:

Sid	Fname	Lname	Branch	Semester	City	DOB	Gender
21BCA001	Vishal	Patel	BCA	2	Anand	18-APR-2002	M
21BCA002	Mahesh	Mandavia	BCA	2		20-JUL-2001	M
20MCA003	Monesh	Viradia	MCA	2	Surat	21-MAR-1998	M
19BCA024	Frida	Pinto	BCA	4	Vasad	18-APR-2002	F
20BCA027	Tanvi	Sharma	BCA	2	Nadiad	04-JAN-2002	F
18BCA045	Vishal	Shah	BCA	4	Anand	08-MAR-2002	M
17MCA060	Shailesh	Macwan	MCA	4		15-APR-2001	M
19BCA100	Rupesh	Misra	BCA	4	Surat	25-AUG-2002	M
17MCA002	Vishal	Mehta	MCA	4	Nadiad	18-OCT-2002	M
17BCA124	Priya	Dave	BCA	4	Vasad	06-FEB-2002	F

2. Table 2: sub_master

Column Name	Datatype and Size	Constraints	Description
Subcode	varchar(10)	Primary Key. Should start with CA	Example:CA119
Subject	Varchar2(40)		subject name
Th_credit	Number(1)	Should be > 0 and <7	Credits in theory portion of the subject



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Pr_credit	Number(1)	Should be > 0 and <4	Credits in practical portion of the subject
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Insert following data in above table:

Subcode	Subject	Th_credit	Pr_credit
CA119	Database Fundamentals	4	3
CA118	Advanced Programming	4	3
CA117	Operating system concepts	4	3
CA221	Data Structures and Algorithms	4	3
CA321	Basics of Mobile Applications		3
CA115	Management Information System	3	0
CA836	Framework and Applications	0	3
CA835	Artificial Intelligence	3	0

3. Table3 : stud_subject

Column Name	Datatype and Size	Constraints	Description
Sid	varchar(10)	Should contain BCA or MCA	Student ID. Example:21BCA001
Subcode	varchar(10)	Should start with CA	Example:CA119

Insert following data in above table:

Sid	Subcode
21BCA001	CA119
21BCA002	CA119
20MCA003	CA119
19BCA024	CA321
20BCA027	CA221
18BCA045	CA836
17MCA060	CA836
19BCA100	CA835
17MCA002	CA835
17BCA124	CA119
21BCA001	CA119
21BCA002	CA119
20MCA003	CA321

Solve Following queries:

1. Display all student details.
2. Display all subject details.
3. List all the students from Anand city.
4. List all the female students.
5. List all the male students.
6. Display name of the students born in April as per the following format:



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Name	DOB
Vishal Patel	18-Apr-2002
Shailesh Macwan	15-Apr-2001

7. List students of BCA branch.
8. List students of MCA branch.
9. List students who have taken admission in year 2017.
10. List BCA students who have taken admission in year 2018.
11. Display details of student with no city.
12. Display gender wise total number of students (Total Male and Total Female).
13. Display details of all students of semester 2.
14. Display age of all the students in following format:

Name	AGE
Vishal Patel	16
Shailesh Macwan	17

15. Display all female students of BCA.
16. Display all male students of MCA.
17. Display all distinct branches
18. Display total credits in all subject as per following format:

Subcode	Subject	Total Credits
CA119	Database Fundamentals	7
CA118	Advanced Programming	7
----	-----	-----

19. Display details of Artificial Intelligence subject.
20. Display only theory subjects.
21. Display only practical subjects.
22. Display all studid and subcode combinations.
23. Display studids who are enrolled for subject code CA836.
24. Display all subject codes of studid 17BCA060
25. Display total number of distinct subject codes.
26. Display total number of distinct studids.
27. Display list of students who have not done course registration.
28. Change the city of roll no 21BCA001 to Mumbai.
29. Insert the city of roll no 21BCA002
30. Increase the size of branch column to varchar2(20).
31. Delete the subject 'Framework and Applications'.

Queries of Expressions/Functions

1. Find eldest student. HINT: use max function.
2. Find youngest student. HINT: use min function.
3. Find total number of students. HINT: use count function.
4. Find total credits in all theory subjects. HINT: use sum function.
5. Find total credits in all practical subjects. HINT: use sum function.
6. Display student's DOB in following format: HINT: use to_char() function.



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Name	DOB
Vishal Patel	April 18, 2002
Shailesh Macwan	April 15, 2001

7. Display student's DOB in following format: HINT: use add_months() function.

Name	DOB
Vishal Patel	18-Oct-2002
Shailesh Macwan	15-Oct-2001

8. Find difference in months between your DOB and today's date.
9. Add following column in the **stud_master created**.

Mobile	Number(10)		
--------	------------	--	--

10. Create following temporary table. Name: **Stud_temp**

Column Name	Datatype and Size	Constraints	Description
Temp_ID	varchar(10)	Not Null	Student ID. Example:21BCA001
Remarks	varchar(20)		

1. Enter the Data in **Stud_temp** table.
2. Display content of the **Stud_temp** Table.
3. Truncate the **Stud_temp table** without explicitly deleting the rows.
4. Again Display content of the **Stud_temp** Table.
5. Drop The Table Stud_temp.

ASSIGNMENT-2

Objectives

- Hands on with Basics of Data Definition Language (DDL)
- To apply JOINS, GROUPBY and Advance SQL queries.

(i) Create tables according to the following definition.

```
CREATE TABLE DEPOSIT (ACTNO VARCHAR2(5) ,CNAME VARCHAR2(18) , BNAME VARCHAR2(18) , AMOUNT NUMBER(8,2) ,ADATE DATE);
```

```
CREATE TABLE BRANCH(BNAME VARCHAR2(18),CITY VARCHAR2(18));
```

```
CREATE TABLE CUSTOMERS(CNAME VARCHAR2(19) ,CITY VARCHAR2(18));
```

```
CREATE TABLE BORROW(LOANNO VARCHAR2(5), CNAME VARCHAR2(18), BNAME VARCHAR2(18), AMOUNT NUMBER (8,2));
```

(ii) Insert the data as shown below.

DEPOSIT

ACTNO	CNAME	BNAME	AMOUNT	ADATE
100	ANIL	VRCE	1000.00	1-MAR-95
101	SUNIL	AJNI	5000.00	4-JAN-96
102	MEHUL	KAROLBAGH	3500.00	17-NOV-95
104	MADHURI	CHANDI	1200.00	17-DEC-95
105	PRMOD	M.G.ROAD	3000.00	27-MAR-96
106	SANDIP	ANDHERI	2000.00	31-MAR-96
107	SHIVANI	VIRAR	1000.00	5-SEP-95
108	KRANTI	NEHRU PLACE	5000.00	2-JUL-95
109	MINU	POWAI	7000.00	10-AUG-95

BRANCH

VRCE	NAGPUR
AJNI	NAGPUR
KAROLBAGH	DELHI
CHANDI	DELHI
DHARAMPETH	NAGPUR
M.G.ROAD	BANGLORE
ANDHERI	BOMBAY
VIRAR	BOMBAY
NEHRU PLACE	DELHI
POWAI	BOMBAY

CUSTOMERS

ANIL	CALCUTTA
SUNIL	DELHI
MEHUL	BARODA
MANDAR	PATNA



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MADHURI	NAGPUR
PRAMOD	NAGPUR
SANDIP	SURAT
SHIVANI	BOMBAY
KRANTI	BOMBAY
NAREN	BOMBAY

BORROW

LOANNO	CNAME	BNAME	AMOUNT
201	ANIL	VRCE	1000.00
206	MEHUL	AJNI	5000.00
311	SUNIL	DHARAMPETH	3000.00
321	MADHURI	ANDHERI	2000.00
375	PRMOD	VIRAR	8000.00
481	KRANTI	NEHRU PLACE	3000.00

From the above given tables perform the following queries:

- (1) Describe deposit, branch.
- (2) Describe borrow, customers.
- (3) List all data from table DEPOSIT.
- (4) List all data from table BORROW.
- (5) List all data from table CUSTOMERS.
- (6) List all data from table BRANCH.
- (7) Give account no and amount of depositors.
- (8) Give name of depositors having amount greater than 4000.
- (9) Give name of customers who opened account after date '1-12-96'.

2. Create table and insert sample data in tables.

Create Table Job (job_id, job_title, min_sal, max_sal)

COLUMN NAME	DATA TYPE
job_id	Varchar2(15)
job_title	Varchar2(30)
min_sal	Number(7,2)
max_sal	Number(7,2)

Create table Employee (emp_no, emp_name, emp_sal, emp_comm, dept_no)

COLUMN NAME	DATA TYPE
emp_no	Number(3)
emp_name	Varchar2(30)
emp_sal	Number(8,2)
emp_comm	Number(6,1)
dept_no	Number(3)

Create table deposit(a_no,cname,bname,amount,a_date).

COLUMN NAME	DATA TYPE
-------------	-----------

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a_no	Varchar2(5)
cname	Varchar2(15)
bname	Varchar2(10)
amount	Number(7,2)
a_date	Date

Create table borrow(loanno,cname,bname,amount).

COLUMN NAME	DATA TYPE
loanno	Varchar2(5)
cname	Varchar2(15)
bname	Varchar2(10)
amount	Varchar2(7,2)

Insert following values in the table **Employee**.

emp_n	emp_name	emp_sal	emp_comm	dept_no
101	Smith	800		20
102	Snehal	1600	300	25
103	Adama	1100	0	20
104	Aman	3000		15
105	Anita	5000	50,000	10
106	Sneha	2450	24,500	10
107	Anamika	2975		30

Insert following values in the table **job**.

job_id	job_name	min_sal	max_sal
IT_PROG	Programmer	4000	10000
MK_MGR	Marketing manager	9000	15000
FI_MGR	Finance manager	8200	12000
FI_ACC	Account	4200	9000
LEC	Lecturer	6000	17000
COMP_OP	Computer Operator	1500	3000

Insert following values in the table **deposit**.

A_no	cname	Bname	Amount	date
101	Anil	andheri	7000	01-jan-06
102	sunil	virar	5000	15-jul-06
103	jay	villeparle	6500	12-mar-06
104	vijay	andheri	8000	17-sep-06



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105	keyur	dadar	7500	19-nov-06
106	mayur	borivali	5500	21-dec-06

3. Displaying data from Multiple Tables (join)

- (1) Give details of customers ANIL.
- (2) Give name of customer who are borrowers and depositors and having living city nagpur
- (3) Give city as their city name of customers having same living branch.
- (4) Write a query to display the last name, department number, and department name for all employees.
- (5) Create a unique listing of all jobs that are in department 30. Include the location of the department in the output
- (6) Write a query to display the employee name, department number, and department name for all employees who work in NEW YORK.
- (7) Display the employee last name and employee number along with their manager's last name and manager number. Label the columns Employee, Emp#, Manager, and Mgr#, respectively.
- (8) Create a query to display the name and hire date of any employee hired after employee SCOTT.

4. To apply the concept of Aggregating Data using Group functions.

- (1) List total deposit of customer having account date after 1-jan-96.
- (2) List total deposit of customers living in city Nagpur.
- (3) List maximum deposit of customers living in bombay.
- (4) Display the highest, lowest, sum, and average salary of all employees. Label the columns Maximum, Minimum, Sum, and Average, respectively. Round your results to the nearest whole number.
- (5) Write a query that displays the difference between the highest and lowest salaries. Label the column DIFFERENCE.
- (6) Create a query that will display the total number of employees and, of that total, the number of employees hired in 1995, 1996, 1997, and 1998
- (7) Find the average salaries for each department without displaying the respective department numbers.
- (8) Write a query to display the total salary being paid to each job title, within each department.
- (9) Find the average salaries > 2000 for each department without displaying the respective department numbers.
- (10) Display the job and total salary for each job with a total salary amount exceeding 3000, in which excludes president and sorts the list by the total salary.
- (11) List the branches having sum of deposit more than 5000 and located in city bombay.

5. To solve queries using the concept of sub query.

- (1) Write a query to display the last name and hire date of any employee in the same department as SCOTT. Exclude SCOTT
- (2) Give name of customers who are depositors having same branch city of mr. sunil.
- (3) Give deposit details and loan details of customer in same city where pramod is living.
- (4) Create a query to display the employee numbers and last names of all employees who earn more than the average salary. Sort the results in ascending order of salary.
- (5) Give names of depositors having same living city as mr. anil and having deposit amount greater than 2000
- (6) Display the last name and salary of every employee who reports to ford.
- (7) Display the department number, name, and job for every employee in the Accounting department.
- (8) List the name of branch having highest number of depositors.
- (9) Give the name of cities where in which the maximum numbers of branches are located.



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(10) Give name of customers living in same city where maximum depositors are located.

6. Manipulating Data

- (1) Give 10% interest to all depositors.
- (2) Give 10% interest to all depositors having branch vrce
- (3) Give 10% interest to all depositors living in nagpur and having branch city bombay.
- (4) Write a query which changes the department number of all employees with empno 7788's job to employee 7844's current department number.
- (5) Transfer 10 Rs from account of anil to sunil if both are having same branch.
- (6) Give 100 Rs more to all depositors if they are maximum depositors in their respective branch.
- (7) Delete depositors of branches having number of customers between 1 to 3.
- (8) Delete deposit of vijay.
- (9) Delete borrower of branches having average loan less than 1000.



ASSIGNMENT-3

Objective:

- To describe constant, variable, variable initialization, and basic operations on the variables in PL/SQL.

1. PL/SQL Block

- 1 Write a PL/SQL block to print 'WELCOME' message.
- 2 Write a PL/SQL block to input and to print your personal detail in the following format using appropriate variable.
Name: <<Name>>
Address: <<Address>>
City: <<City>>
Mobile: <<Mobile Number>>
Birth Date: <<Birth date>>
- 3 Write a PL/SQL block to print to initialize two number variables and print the total of it.
- 4 Write a PL/SQL block to swap the values of two number type of variables. Print the values of variables before and after the swap.
- 5 Write a PL/SQL block to print minimum of two numbers.
- 6 WRITE A PL/SQL BLOCK TO FIND MAX OF THREE NUMBERS.
- 7 WRITE A PL/SQL BLOCK TO PRINT GRADE OF THE STUDENT
INPUT : ROLLNO, MARKS1, MARKS2, MARKS3
CALCULATE TOTAL = MARKS1 + MARKS2 + MARKS3 / 3
BASED ON TOTAL CALCULATE PER
GRADE IS CALCULATED AS PER FORMULA
IF PER >80 GRADE IS AA
IF PER BETWEEN 70 AND 80 GRADE IS A
IF PER BETWEEN 60 TO 70 GRADE IS AB
IF PER BETWEEN 50 TO 60 GRADE IS B
IF PER BETWEEN 40 TO 50 GRADE IS C
IF PER LESS THAN 40 GRADE IS FAIL
- 8 WRITE A PL/SQL BLOCK TO PRINT 10 NUMBERS USING WHILE AND FOR LOOP
- 9 WRITE A PL/SQL BLOCK TO PRINT MULTIPLICATION TABLE
USING WHILE AND FOR LOOP
- 10 WRITE A PL/SQL BLOCK TO PRINT FACTORIAL OF GIVEN NUMBER.



ASSIGNMENT-4

Objective:

- To describe discuss DML statement in PL/SQL with exception handling also.

PL/SQL Implicit Cursor

- 1 Write a PL/SQL block to display name and mobile number from EMP table whose id is 101.
- 2 Write a PL/SQL block to insert 5 numbers in temp table, update the any number by taking input from user, delete any number as per user choice. Also print appropriate message.
- 3 WRITE A PL/SQL BLOCK TO INSERT ODD NUMBERS IN T1 TABLE AND EVEN NUMBERS IN T2 TABLE
WRITE A PL/SQL BLOCK TO DELETE VALUES FROM TABLE T1
WRITE A PL/SQL BLOCK TO DELETE LAST 50 RECORD OUT OF 100 FROM TEMP TABLE
- 4 Create table of patient with fields
PATIENT(pid integer, pnm varchar2(10), age integer)
---> TAKE PATIENT ID AS PRIMARY KEY,
AGE SHOULD BE GREATER THAN ZERO
ADD GENDER FIELD INTO PATIENT TABLE AND UPDATE THE RECORD

Write a PL/SQL block which will print
the patient report in the following format
INPUT PATIENT ID FROM USER AND
COMPARE IT WITH TABLE ID AND DISPLAY ACCORDINGLY

patient id: XXX Patient name: XXX
AGE <20 : PRINT IT
AGE >= 20 : PRINT IT
AGE >20 and AGE <=30 : PRINT IT
AGE > 30 & AGE <=40 : PRINT IT
AGE >40 & AGE <60 : PRINT IT
AGE >=60 : DISPLAY IT

- 5 WRITE A PL/SQL BLOCK TO DO THE FOLLOWING:
INPUT 20 NUMBERS FROM USER
A) FIRST ADD ALL NUMBERS IN T3 TABLE
B) IF NUMBER IS 5,10,15 THEN DO UPDATE T3 (ADD 20 IN)
ELSE DELETE OTHER RECORDS
Print appropriate message for valid and invalid record.



ASSIGNMENT-5

Objective:

- To describe SELECT-into clause – only one output statement in PL/SQL

```
CREATE TABLE DOCTOR(DID INTEGER PRIMARY KEY, DNAME  
VARCHAR2(10))
```

```
CREATE TABLE PATIENT(pid integer PRIMARY KEY,  
patientnm varchar2(10),  
age integer CHECK(AGE > 0),  
GENDER CHAR(1) CHECK(GENDER IN ('M','F','m','f')),  
DID INTEGER REFERENCES DOCTOR)
```

INSERT 5-7 RECORDS IN BOTH TABLE

```
SELECT * FROM DOCTOR  
SELECT * FROM PATIENT
```

1. Write a PL/SQL block to print senior citizen(maximum age of patient).
2. Write a PL/SQL block to print name of patient who is youngest.
3. WRITE A PL/SQL BLOCK TO COUNT TOTAL NUMBER OF RECORDS FROM PATIENT TABLE
4. WRITE A PL/SQL BLOCK TO DISPLAY MESSAGE OF PATIENT
E.G. TAKE INPUT AS PID, EXTRACT AGE AND DO THE FOLLOWING

TAKE AGE FROM TABLE AND BASED ON CONDITION DISPLAY THE RECORD

IF AGE >= 60 THEN DISPLAY- YOU ARE SENIOR CITIZEN

IF AGE >40 AND AGE <60 DISPLAY- YOU ARE MATURE CITIZEN

IF AGE >20 AND AGE <=40 DISPLAY- YOU ARE YOUTH CITIZEN

IF AGE >14 AND AGE <= 20 DISPLAY - YOU ARE TEENAGER

IF AGE <=14 DISPLAY YOU ARE CHILD

- 5 WRITE A PL/SQL BLOCK TO PRINT DEPT NAME OF DEPT ALONG WITH
EMPLOYEE NAME WHOSE DEPTNO = 2



ASSIGNMENT-6

Objective:

- To describe Explicit cursor- ALL records from table or selected records from table and manipulate data based on logic.

- 1 Assume that EMP and DEPT table with proper records exists in your database.

EMP(empid, fname, mname, lname, gender, salary, dob, designation, phoneno, deptid)

DEPT(deptid, dname, max_capacity)

Write a PL/SQL block to display all the records.
(use explicit cursor and cursor-For loop)

- 2 Write a PL/SQL block to display name, phone-no and designation of female employees.
- 3 Write a PL/SQL block to display highest paid first 5 employees.
- 4 ADD bonus field in EMP table.

Write a PL/SQL block to calculate bonus based on condition. After calculation of bonus of all the employees, display on screen as well as update it.

IF salary greater than 50k then bonus is 10% of salary

IF salary greater than 25k and less than 50k bonus is 15% of salary

If salary less than 25k then bonus is 20% of salary.

- 5 Assume that you have EMP and DEPT table with proper records:

Find out the name of employee who has experienced in the Electrical department more than 5 years.

- 6 Assume that you have EMP and DEPT table with proper records:

Write a PL-SQL block that will print the following report:

Id	NAme	Gender	Phone-no	Designation	Deptid	Deptname
1	Mahendra	M	9999222234	Manager	101	MCA
2	Virat	M	9873627789	Sales Manager	102	Civil
3	Rohit	M	8762349087	Clerk	101	MCA
4	Renu	F	7896734567	Accountant	103	Electrical

Total Records Printed : 4



ASSIGNMENT-7

Objective:

- To discuss Stored Procedure in PL/SQL from block or direct.

Assume that EMP and DEPT table with proper records exists in your database.

EMP(empid, fname, mname, lname, gender, salary, dob, designation, phoneno, deptid)

DEPT(deptid, dname, max_capacity)

- 1 Write a Procedure which display average salary of employee.
- 2 Write a Procedure which pass two parameters, calculate sum of two values and display it in procedure.
- 3 Write a Procedure which will display total no. of employees whose salary greater than 10000 and execute it.
- 4 Write a Procedure which will display mobile number of employee whose id is given by user. (pass as parameter)
- 5 Create a stored procedure that takes and input parameter as department- id and generate an output from the procedure as number of employees who are working in that department.
- 6 Write a procedure for following task:
When new employee joins in the institute, check if empid is present in the institute or not and the no. of already registered employees *not* exceeding the max capacity for that institute.
- 7 Write a procedure to display top five highest paid employees whose designation is Officer.
- 8 Write a procedure to display name of employee from emp table in a given format: (first name, middle name, last name), use CASE statement to have 4 options.
For example,
 - Display name in the format “ RLP “
 - Display name in the format “ R.L.Patel “
 - Display name in the format “ Rambhai L. Patel “
 - Display name in the format “ Rambhai Laxmanbhai Patel”
- 9 Enter the employee id from user. Check whether the name exists in table for corresponding id. If yes, Write a PL/SQL procedure to find total no of vowels from employee name. If no, display appropriate message.
- 10 To vote for Election, write a Procedure which will display the following details:
Eligible to vote(>18 years) || Not Eligible (less than 18 years)
Id & name of employee || id & Name of Employee



ASSIGNMENT-8

Objective:

- To discuss Stored Function and local function in PL/SQL from block.

- 1 Write a function to take two numbers from user and print summation of it.
- 2 Write a function whether given number is odd or even.
- 3 Write a function that returns total number of incomplete jobs, using table
JOB(jobid, type_of_job, status)
- 4 Write a function to which a birthdate is passed and it returns the age in terms of years as of today and a procedure/block which calls this function to display.
- 5 Write a function which displays the number of items whose weight fall between a given range for a particular color using table
ITEM(itemno, name, color, weight)
- 6 Create a function to return count of all employees who works in MCA department. This function is called in procedure which displays the output in proper format.

- 7 ISSUE (ROLLNO, BOOKNO, ISSUE_DATE, RETURN_DATE)

After calculating the fine store the required information in the FINE table so that a report can later be printed out.

FINE (ROLLNO, BOOKNO, ISSUE_DATE, RETURN_DATE, TOT_DAYS, FINE)

Write a procedure or function for the given a table, check which students have to pay fine and the amount of fine to be paid. A student can keep the book for 15 days. If the number of days has exceeded 15 then, the fine is calculated as follows:

Up to 7 days : 50 paise per day

8 – 15 days : Re 1 per day (to be counted from the 8th day onwards)

More than 15 : Re 1.5 per day (to be counted from the 16th day onwards)

- 8 NEWS_PAPER(news_id, name_of_paper, language, month_frequency, price)

Write a procedure that will display name of paper which is of lowest price.

Write a function which will count total number of newspaper exists language wise.

(English language : 999 papers

Gujarati language : 999 papers)

Hindi language : 999 papers)



ASSIGNMENT-9

Objective:

- To discuss Database Trigger in PL/SQL.

Assume above EMP and DEPT table exists with records.

- 1 Write a code which demonstrate simple trigger.
- 2 Create a trigger on EMP table that tracks the UPDATE on it. Each record being updated on it shall be tracked and kept as backup in EMP_TRAN table
- 3 Create a trigger on EMP table that tracks the DELETE on it. Every record being deleted from it shall be kept in EMP_TRAN_BACKUP table.
- 4 Create a trigger on EMP table that tracks UPDATE being performed on SAL column of the table. If the UPDATE is being performed between 2.30 pm to 3.30 pm then copy the effect on EMP_UPDATE_TRACK table
- 5 Write a trigger which restrict to enter or update the transaction on Saturday or Sunday
OR
Write a PL\SQL trigger which will not allow to insert records on saturday and sunday.
- 6 Write a trigger which restrict user to enter name in EMP table either starting with 'X' or is less than 2 letters
- 7 ADD Blood group in EMP table.

Write a trigger which only accept blood_gr in range(with +ve and -ve blood gr) of A ,B, AB, O.
- 8 Write a trigger in which accepts empid from user and check whether sal>65000 , don't allow to insert or update emp table.
- 9 Create a trigger which ensures that Emp name that to be inserted is in capital.
- 10 Write a trigger which deletes every record from result table if student table is empty.
student(studno,fnm,mnm,lnm,degree,bdth);
result(studno,math,eng ,ss,sci)
- 11 Write a trigger which accepts student records whose age is between 18 to 30.
- 12 Consider the table: (Eno, Ename, Basic, Hra, Da, Total)
Assume the following :
 - * Hra is given in Rupees
 - * DA is given as percentage as basic
 - * Total salary will be Basic + da*basic/100 + hra(i) Write a trigger so that the record on insertion contains basic salary in the range of 5000 to 18000;
DA in the range 16% to 32%; and HRA not more than 3000.



ASSIGNMENT-10

Objective:

- To discuss PL/SQL Parameterized cursor, Exceptional handling and package.

- 1 Item (Item code, item name, unit price)
Bill (bill number, bill date)
Bill_item (bill number, item code, number of units purchased)

Do the following:

- (a) For a given bill number, print the bill in the following format.

Bill no.:

Bill Date:

Sr. No.	Item code	Item name	Units purchased	unit price	total price
###	xxxxx	xxxxxxxxxxxxxxxxxxxxx	#####	####.##	#####.##

Bill Amount: #####.##

- (b) Add one more attribute named bill_amount in the Bill table. Update the Bill table to store bill amount.

- 2 **PATIENT(pid, pname, age, gender, doctor-id)**

Prepare following report for a Doctor information related to total patient age wise in his hospital.

Gender	Patient in Age Group					Total
	<u>1-12</u>	<u>13-25</u>	<u>25-50</u>	<u>50-100</u>	<u>Above 100</u>	
----	----	----	----	----	----	----
Total	-----	-----	-----	-----	-----	-----

Patient (pid, pame, age, gender , doctor-id)

Appoint(pcode, app_date, time , doctor_id)

Doctor(doctor_id, name, specialization)

Prepare a report from above table taking date, shift, , doctor_id as parameters.

Shift is Morning if time are from 9.00 Am to 1.00 pm and

Shift is Evening if time are from 5.00 to 9.30 pm.

APPOINTMENT SCHEDULE

Date :

Doctor name :

Shift :

Specialization :

Time

Patient name

Age

9.00

9.10

9.20

.....



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Total no. of patients : _____

- 3 ALBUM(album_no, album_name, album_type, release_date, singer_name, no_songs)

Write a procedure which takes parameter as singer name and print the following album details:

Singer name : _____

Album no	Album name	Type	Release Date	Number of Songs
99	Xx	Xx	Xx	99
99	Xx	Xx	Xx	99

- 4 Create table and insert proper data.

Tour_Details(tour_no, start_date, group_size, source, destination, vehicle_no, driver_name)

Write a PL/SQL block which will print tour details, a driver is going to take it. (pass driver_no as parameter)

Driver Name : _____ Vehicle number : _____

Tour Details

Source	Destination	Group size	Start date	Total days
Xxx	Xxx	999	Xxx	999

- 5 CUST(cno, accno, name, add, city, phone, email)
ACCOUNT(accno, amount)

Write a stored procedure for banking purpose which perform following Task :

1. Create a new account (insert)
2. Deposit funds (update)
3. Check account balance (view balance)
4. Close the account (delete)

- 6 Create table and insert proper data.

TRAIN_MAST(train_no, train_name, arrival_time, departure_time, source, destination)

Write a procedure which will print all train details going from Baroda to Bombay
Write a function which will print arrival time and departure time for a given train. (pass train no as a parameter)



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7 Given hospital schema

Patient(pat_code, pname, birthdate, gender)

Dentist(dcode, dname, qualification, area, city, experience)

Treatment(dcode, pat_code, treatment_date, treat_details)

Write a procedure or Function to do the following queries: (USE Switch case)

1. Count no. of dentist from 'lalbaug' area in baroda city.
2. Display details of dentist with 'MDS' as qualification.
3. Display details of treatment given to patient no.'P101'.
4. Display total number of patient who came for treatment as on '12-Feb-2013'.
5. Display area wise list of dentist who has maximum experience.
6. Display details of youngest dentist.
7. Display dentist details along with their patient.
8. Display details of all female dentist.
9. Count total number of male dentist who live in 'ahmedabad'.
10. Display patient name without duplicate who come for treatment.
11. Display details of patient who has not given any treatment.

8 Consider the EMPLOYEE schema as :

EMPMAS(empno, name, pfno, empbasic, deptno, designation)

HOLIDAY(year, month, holidays, weekoff)

EMPTRAN(empno, month, year, presence, loan(amt))

BANK(accno, empno, bank name, bank branch)

Rules : HRA = 15% of basic

DA = 50% of basic

Medical = 100

PF = 8.33% of basic

Salary is given for (attendance + holidays + weekoff) days.

Print salary slip in foll. Format.

SALARY SLIP of Month : _____ Year _____			
Accno :	Name of EMP	Designation	Dept
Bank name : _____ Branch : _____ Dt of Join _____			
Presence : _____		Absent days : _____ Sal days : _____	
EARNINGS :		DEDUCTIONS	
BASIC :		LOAN :	
DA :		PF :	
HRA :		Prof tax : 100	
MEDICAL : 100			



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TOTAL earnings: _____

TOTAL Deductions: _____

NET PAY : _____

9 Write a code which demonstrate user defined exception.

10 Write a code which demonstrate package in PL/SQL.

=====THERE IS BEGINNING IN THE END !!!=====



Assignment-I

Data Pre-processing Tasks – Part I

1. Get an overview of the dataset
 1. Read the .csv file
 2. Read .csv file and consider “na”, “N/A”, “Not Available” as set of missing values
 3. Print the entire dataset in Pandas data frame
 4. Print datatype of all the columns of dataset
2. Get a statistical summary of the dataset
 1. Display all the statistical indicators of your dataset
 2. Print the name of all unique columns
 3. Print the count of all unique columns
3. Get a subset of the entire dataset
 1. Create a new Pandas dataframe and copy ith rows and jth columns of your dataset
4. Modify the dataset
5. Identify and deal with missing values
 1. Display all the tuples with status of missing values
 2. Count the number of missing values in dataset of columns
 3. Print the columns having missing value
 4. Drop all the rows having atleast one missing values in the same Pandas dataframe
 5. Drop all the rows where all the values are missing values in the same Pandas dataframe
 6. Drop all the columns having atleast one missing values using different Pandas dataframe
 7. Drop all the columns where all the values are missing values using different Pandas dataframe
 8. Fill the missing value with constant
 9. Fill the missing value with mean, median and mode
 10. Fill the missing value with forward fill, backward fill and interpolate



Assignment-II

Data Pre-processing Tasks – Part II

1. Binning using Python
 1. Implement Binning using cut and qcut methods.
 2. Also, transform the bins values.

2. Outlier detection and removal
 1. Detect the outlier using visualization method
 2. Detect the outlier using statistical method
 3. Treat the outliers



Assignment-III

Classification and Prediction

1. Read the seeds.csv file
2. Perform all data pre-processing techniques.
3. Perform one-hot encoding and normalization.
4. Apply decision tree algorithm and display classification report.
5. Apply naïve-byes algorithm and display classification report.

Assignment-IV

Clustering

1. Read the Wholesale_customers_data.csv file.
2. Perform all data pre-processing techniques.
3. Perform one-hot encoding and normalization.
4. Apply k-means clustering algorithm.
5. Apply hierarchical clustering algorithm.